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Enhancement of Fermentation Performance in the Anaerobic Co-Digestion of Chicken Manure and Corn Straw under Biogas Slurry Reflux via Air Stripping of the Digestate

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Abstract: Ammonium inhibition is a key limiting factor for anaerobic digestion when using chicken manure as the main substrate, especially in a digestion system with biogas slurry reflux. Air stripping is usually used as a recycled biogas slurry treatment. In this study, we carried out the anaerobic co-digestion of chicken manure and corn straw. The fermentation performance was investigated with and without air stripping at different biogas slurry reflux ratios and with an increasing organic loading rate. The results show that air stripping enhanced biogas production, system stability, and volatile solid removal efficiency via the mitigation control of ammonium inhibition. The total ammonium nitrogen in the digesters with air stripping was 20.24–46.40% lower than in those without air stripping. The highest specific biogas production and volatile solid removal efficiency values were obtained in the digesters at an organic loading rate of 3.3 g volatile solid (VS)/(L·d) and a reflux ratio of 75% with air stripping, reaching 480.43 mL/gVS_{add} and 63.36%, respectively. Moreover, air stripping also improved the organic loading rate and reflux ratio. Stable operation was achieved at an organic loading rate of 5.3 gVS/(L·d) and a reflux ratio of 75%, with specific biogas production of 392.35 mL/gVS_{add} and a volatile solid removal efficiency of 50.33%. The fermentation performance deteriorated when the organic loading rate was increased to 8.0 gVS/(L·d) at a reflux ratio of 75%, even when air stripping was conducted, indicating that a slighter lower reflux ratio (50%) could be more feasible at a higher organic loading rate (8.0 gVS/(L·d). Additionally, the methanogen community structure varied according to the use of air stripping, with a shift in the methanogenic pathway from hydrogenotrophic to acetoclastic methanogens. Overall, our findings support the adoption of air stripping for ammonium mitigation in anaerobic digestion with biogas slurry reflux.

Keywords: anaerobic digestion; biogas slurry reflux; chicken manure; corn straw; air stripping; ammonium inhibition; organic loading rate

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1. Introduction

Anaerobic digestion is an effective technology for safely treating agricultural waste and producing bio-methane and biogas slurry, which can be used as bio-fertilizers. These two positive qualities have seen anaerobic digestion gain particular importance in environmental protection and the production of renewable energy in rural areas [1]. In recent years, biogas plants fed agricultural waste have been developed rapidly in China. According to statistics, there were more than 100 thousand biogas plants fed agricultural waste, with

a total volume of 20 million m^3, at the end of 2017, compared with 17 thousand, with a total volume of 2.84 million m^3, at the end of 2007 [2,3]. With the increase in biogas plants, increasing amounts of biogas slurry are being generated, accompanied by increasing environmental pollution. The rational and economical use of biogas slurry has become key to the sustainable development of biogas engineering [4,5].

Biogas slurry has traditionally been used as a bio-fertilizer due to its high nutrient content. However, the field application of biogas slurry is challenging for most biogas plant operators in China because the surrounding farmland is deficient and scattered. Furthermore, biogas slurry is produced year-round, but the field application of biogas slurry is seasonal according to fertilizer demand; therefore, it needs to be stored when it cannot be applied to the field. The storage of biogas slurry emits a large amount of greenhouse gasses into the atmosphere, leading to secondary pollution [6–8].

Biogas slurry reflux technology is widely applied by most biogas plants in China due to its potential for biogas slurry discharge reduction, water saving, and waste heat recovery. In addition, it can return microorganisms and organic substances to the anaerobic digestion system, thereby improving biogas production. However, ammonium nitrogen, volatile fatty acids, and other refractory compounds in the anaerobic digestion system accumulate during the long-term reflux of biogas slurry, leading to biogas production inhibition and system instability [9,10]. Ammonium inhibition is the main problem in the anaerobic digestion of chicken manure, which has a high content of organic nitrogen. The problem of ammonium inhibition is more serious in the case of biogas slurry reflux because biogas reflux requires the continuous recirculation of the discharged biogas slurry into the anaerobic digestion system, which introduces more ammonium nitrogen into the system [11,12].

Many methods to avoid ammonium inhibition by decreasing or removing ammonium have been investigated, such as co-digestion, dilution, gas stripping, vacuum stripping, and struvite precipitation. Among them, the most common method is the co-digestion of chicken manure with carbon-rich substrates such as crop straw [13]. However, our previous study found that ammonium inhibition still occurred when 50% or more of the discharged biogas slurry was recycled, even though corn straw was used as the co-digestion substrate for chicken manure. The ammonium mitigation process, when using carbon-rich substrates for anaerobic co-digestion, is slow and must be utilized before the anaerobic system is completely inhibited [14]. Once the biogas production has drastically reduced or even stopped, it is difficult for the system to recover from the ammonium inhibition state [15]. Therefore, it is more effective for ammonium mitigation to remove ammonia from the biogas slurry via physical or chemical methods before its re-utilization [16,17]. Air stripping, an effective physical method, has been successfully applied in practical biogas production engineering due to its simple operation, low land occupation, and high ammonium removal efficiency [11,18]. Previous studies have reported the positive effects of air stripping on biogas production and anaerobic digestion system stability for ammonium mitigation before biogas slurry recirculation. Wu et al. [11] reported that air stripping could be integrated for the ammonium mitigation of recycled liquid digested slurry. With air stripping, ammonium was reduced from 8000 mg/L to 3000 mg/L, and biogas production increased from 0.8 ± 0.1 L/(L·d) to 1.4 ± 0.1 L/(L·d). The mono-digestion of chicken manure was successfully carried out at high ammonium concentrations with recycled air-stripped liquid digested slurry in the research by Nie et al. [12]. The organic loading rate (OLR) of 5.3 g volatile solid (VS)/(L·d) was achieved with a specific biogas yield of 0.39 L/gVS. When the OLR was increased to 6.0 gVS/(L·d), stable operation was achieved with a specific biogas yield of 0.27 L/gVS. However, the enhancement of the anaerobic digestion process by recirculating the air-stripped biogas slurry with different reflux ratios has not been reported. Moreover, most previous studies focused on the effects of air-stripped biogas slurry recirculation on biogas production; the enhancement effect of the OLR and the reflux ratio remains unknown. Methanogens are one of the three groups of microbes (hydrolytic, acid-forming, and methanogenic bacteria) in anaerobic digesters, and

they are considered more susceptible to ammonium inhibition [19]. Different ammonium nitrogen loads may alter the methanogenic community structure and methane production pathway [20].

In this study, we used air stripping as a recycled biogas slurry treatment (1) to assess its positive effects on biogas production, fermentation characteristics, and volatile solid (VS) removal efficiency; (2) to analyze whether air-stripped biogas slurry recirculation can feasibly enhance the OLR and reflux ratio; and (3) to determine the dynamic changes in the methanogen community in different anaerobic digesters when increasing the OLR.

2. Materials and Methods

2.1. Fermentation Materials and Inoculum

Corn straw obtained from the experimental field of Lifang Organic Dryland Agriculture Experimental Base of Shanxi Agricultural University (Jinzhong, China) was crushed to 2–3 mm size after natural ventilation and drying. Chicken manure was collected from Taigu Honghao Breeding Cooperation, located in Jinzhong, China. The inoculum (Biogas Technology Laboratory of Shanxi Agricultural University, Taiyuan, China) to start the digesters was collected from the fermentation residue of chicken manure and corn straw. After collection, the chicken manure and inoculum were stored in a refrigerator at 4 °C. The characteristics of the fermentation materials and inoculum are given in Table 1.

Table 1. Characteristics of the fermentation materials and inoculum.

Materials	TS/%	VS/%	C/N	pH
Chicken manure	50.40	40.93	15.31	7.56
Corn straw	90.46	80.92	42.07	ND
Inoculum	7.73	5.37	ND	7.23

Note: ND: no detection; TSs: total solids; VSs: volatile solids; C/N: the ratio of total carbon to total nitrogen.

2.2. Experimental Devices

Four laboratory-scale continuously stirred tank reactors (CSTRs) were operated in a semi-continuous mode at a temperature of 35 °C. The reactors were manufactured by the Biogas Technology Laboratory of Shanxi Agricultural University (Taiyuan, China) [19]. Their design is illustrated in Figure 1. They included a cylindrical plexiglass tank with an inner diameter of 20 cm and a height of 32 cm. The total volume of the reactors was 10 L and the working volume was 7.5 L. The reactors also included a mixing motor, an inlet, an outlet, and a heating jacket for water bath heating. A mechanical stirrer provided batch mixing with a stirring intensity of 60 r/min, stirring duration of 20 min, and stirring cycle of 6 h.

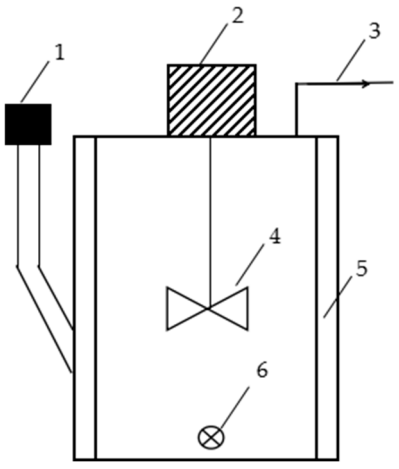


Figure 1. Schematic diagram of the anaerobic digesters. 1: Feeding inlet; 2: mixing motor; 3. biogas outlet; 4: stirrer; 5: heating jacket; 6: feeding outlet.

2.3. Experimental Design

The same operation was carried out in the four digesters to start the reactors. In the start-up experimental period, chicken manure (538.73 g), corn straw (181.67 g), inoculum (2935.06 g), and water (3844.55 g) were placed into the reactor. Then, 357 mL of fresh substrate mixture with 7% VS (6:4 chicken manure and corn straw) was fed into the reactor and an equivalent volume of digestate was discharged every day. The hydraulic retention time (HRT) was 21 days, and the OLR was 3.3 gVS/(L·d) (calculated by the working volume, the same below). Once the reactors were running under stable conditions, the experiment entered the study period.

In the study period, biogas slurry recirculation was carried out in each reactor. The reflux ratio of R1 and R2 was 50%, and of R3 and R4 was 75%. The reflux ratio was calculated using Equation (1).

$$r\% = \frac{V_0}{V_0 + V_1} \times 100 \quad (1)$$

where r represents the biogas slurry recirculation ratio, V_0 represents the daily recycled biogas slurry volume, and V_1 represents the daily feeding water volume. A total of 357 mL of digestate was discharged from each reactor each day. After filtration through a screen with a diameter of 2 mm, the liquid fraction of the discharged digestate was treated in an air stripping device to remove ammonia, then used to prepare the feeding substrate mixture according to the specified reflux ratio of R1 (50%) and R3 (75%). The air stripping process consisted of first raising the pH of the filtered liquid to 10.5 by adding $Ca(OH)_2$, then stripping the filtered liquid with air for 2.5 h, with an air flow ratio (the ratio of air flow volume to biogas slurry volume) of 4000. For digesters R2 and R4, the filtered liquid was directly recycled without air stripping, while the other operation conditions remained the same as for R1 and R3. The anaerobic digestion was run in three stages of 42 d each, for a total of 126 d in each digester. The OLR for each digester was 3.3 gVS/(L·d) in stage I (0~42 d) and was increased gradually to 5.3 and 8.0 gVS/(L·d) in stage II (43~84 d) and III (85~126 d), respectively. The operation conditions of each digester are given in Table 2.

Table 2. Operation conditions of the four digesters.

Digester	HRT/d	Reflux Ratio/%	Air Stripping	OLR/gVS/(L·d)		
				Stage I 0~42 d	Stage II 43~84 d	Stage III 85~126 d
R1	21	50	Y	3.3	5.3	8.0
R2	21	50	N	3.3	5.3	8.0
R3	21	75	Y	3.3	5.3	8.0
R4	21	75	N	3.3	5.3	8.0

Note: Y indicates that the liquid fraction of the discharged digestate was treated using air stripping to remove ammonia; N indicates it was untreated.

2.4. Analytical Methods and Calculations

(1) Biogas

The daily biogas production was measured periodically by the water displacement method.

(2) Materials and digestate

Total solids (TSs), VS, total carbon, and total nitrogen were analyzed using standard methods [21]. The pH value was determined using a pH meter (CyberScan PC 300, San Francisco, CA, USA). For the analysis of total ammonium nitrogen (TAN) and volatile fatty acid (VFAs), the digestate was centrifuged at 8000 r/min for 20 min, and the supernatant was analyzed. The mass concentration of VFAs was determined using the colorimetric method, as described by Su [22], using a spectrophotometer (UV-5200PC, Shanghai, China). The TAN mass concentration was determined by the distillation–neutralization method

according to the standard [23], and the free ammonia (FAN) mass concentration was calculated using Equation (2).

$$\mathrm{FAN} = \frac{\mathrm{TAN}}{1 + 10^{(\mathrm{pKa} - \mathrm{pH})}} \quad (2)$$

where Ka is the dissociation constant of NH_3, pKa = 8.9 at 35 °C.

The VS removal efficiency was calculated using the equation reported by Nie et al. [12] (Equation (3)):

$$\omega\% = \left[1 - \frac{\mathrm{VS_{out}} \times (100 - \mathrm{VS_{in}})}{\mathrm{VS_{in}} \times (100 - \mathrm{VS_{out}})}\right] \times 100 \quad (3)$$

where ω represents the VS removal efficiency. VS_{in} and VS_{out} represent the VS concentration of the daily added feed mixture % and discharged digestate %, respectively.

In this study, the pH, TAN, and VFAs were analyzed every seven days. Biogas production and VS were measured daily.

(3) Microbial analysis

Digestate samples from the four digesters were collected at different OLRs on day 42, 84, and 126 for the methanogenic microorganism analysis. DNA was extracted from the samples using a TGuide S96 HiPure Soil DNA extraction kit (Tiangen Biotech, Beijing, China). The V3 and V4 region of the methanogenic archaea were amplified using the double-ended primers Arch349 (5′-GYGCASCAGKCGMGAAW-3′) and Arch806R (5′-GGACTACVSGGGTATCTAAT-3′). The amplified and purified samples were sequenced using the Illumina Novaseq sequencing platform (Biomarker Co., LTD., Beijing, China).

2.5. Data Processing

Microsoft Excel 2007 was used for data analyses and graphics drawing.

3. Results and Discussion

3.1. Biogas Production

The volumetric biogas production rate (VBPR) and specific biogas production (SBP) of the four digesters are shown in Figure 2. Overall, the VBPR and SBP of R1 and R3 were higher than that of R2 and R4. R1–R4 showed the same trend, wherein the VBPR increased with the increase in the OLR from 3.3 to 5.3 gVS/(L·d), then decreased when the OLR increased to 8.0 gVS/(L·d). However, the improvement in the OLR caused a decrease in the SBP in all digesters. The highest VBPR was obtained in R3 at the OLR of 5.3 gVS/(L·d), reaching 2.09 L/(L·d), which was 67.20% higher than R4 (without air stripping at the same reflux ratio of 75%), 19.43% higher than R1 (with air stripping at the reflux ratio of 50%), and 58.33% higher than R2 (without air stripping at the reflux ratio of 50%). As the OLR increased to 8.0 gVS/(L·d), the VBPR in R2–R4 dropped sharply, with an average value of 0.71, 0.86, and 0.56 L/(L·d), which was 25.16%, 46.20%, and 35.69% lower than that at the OLR of 3.3 gVS/(L·d). R1 maintained a VBPR level (1.74 L/(L·d)) as high as that at the OLR of 5.3 gVS/(L·d) (1.75 L/(L·d)). The highest SBP (480.43 mL/gVS_{add}) was in R3 at the OLR of 3.3 gVS/(L·d), followed by R1 (357.37 mL/gVS_{add}). The SBP in R1–R4 decreased, with an average value of 328.44, 246.85, 392.35, and 224.38 mL/gVS_{add} during stage II. A relatively steady decreasing trend in the SBP was observed in digester R1, with an average value of 217.64 mL/gVS_{add} as the OLR increased to 8.0 gVS/(L·d), while sharp drops of the SBP were seen in R2–R4 at the OLR of 8.0 gVS/(L·d), with average values of 93.45, 95.80, and 69.44 mL/gVS_{add}. R3 had the largest reduction in efficiency of 77.58% and 72.55% compared with the OLR of 3.3 and 5.3 gVS/(L·d), respectively. Although air stripping was conducted, biogas production inhibition still occurred as the OLR increased to a higher level (8.0 gVS/(L·d)) under a high reflux ratio (75%), indicating that a lower reflux ratio (50%) should be adopted at a higher OLR.

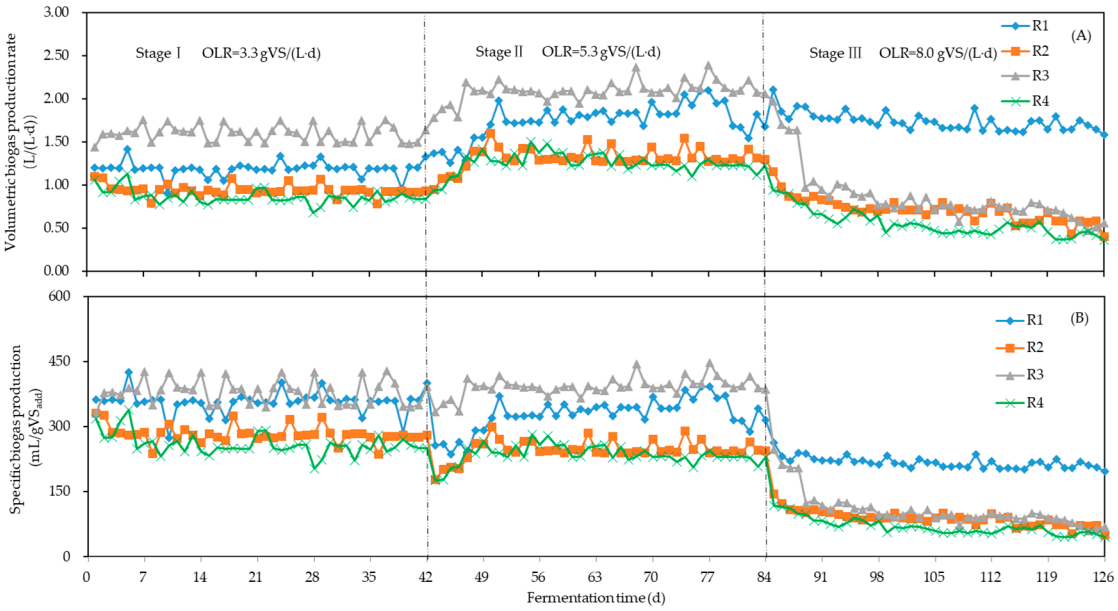


Figure 2. Comparisons of the volumetric biogas production rate (**A**) and specific biogas production (**B**) between digesters with and without air stripping.

3.2. Changes in TAN, FAN, pH, and VFAs

TAN, FAN, pH, and VFAs are important chemical parameters during the fermentation process and are considered to be indexes of the fermentation stability [24,25].

3.2.1. TAN and FAN

It can be clearly seen from Figure 3A that, overall, the TAN values of R1 and R3 (with air stripping) were 20.24–46.40% lower than R2 and R4 (without air stripping). During stage I, the levels of TAN and FAN in R1–R3 were lower than most previously reported inhibition thresholds (TAN 3000 mg/L, FAN 200 mg/L) [26,27], while the concentrations of TAN and FAN in R4 fluctuated around 3000 and 200 mg/L, respectively. As the OLR increased to 5.3 gVS/(L·d) in stage II, the TAN and FAN values in all digesters increased. The values of TAN and FAN in R2 and R4, without air stripping, far exceeded the inhibition threshold at 3000 mg/L and 200 mg/L. The average values of FAN in R2 and R4 reached 452.40 and 668.79 mg/L, which were more than double and triple the value of 200 mg/L, respectively. However, the levels of FAN in R1 and R3 were reduced, with average values of 157.33 and 222.10 mg/L, when air stripping was adopted. FAN has a stronger inhibitory effect on methanogens compared to NH_4^+ because it is freely permeable through microbial cell membranes and inhibits enzyme activity associated with methanogens [15,28]. High values of TAN and FAN resulted in a reduction in VBPR and SBP in R2 and R4 (Figure 2). During stage III, the TAN concentrations in RI–R4 further increased at an OLR of up to 8.0 gVS/(L·d), with average values of 2642.18, 4889.58, 3549.05, and 6212.62 mg/L, respectively. The FAN value in R1 also increased to an average value of 305.71 mg/L. It should be noted that the FAN concentrations in R2–R4 exhibited a downward trend, with average values of 149.89, 108.22, and 61.44 mg/L, as the OLR increased to 8.0 gVS/(L·d), whereas the VBPR and SBP values dropped sharply (Figure 2). This was because, although TAN increased in R2–R4, the pH values decreased (Figure 4A) due to the accumulation of VFAs (Figure 4B). According to Equation (2), FAN decreases with the decrease in pH. The interaction of FAN, pH, and VFAs placed the digesters in an "inhibited steady state". Previous studies have shown that the "inhibited steady state" is the state in which ammonium inhibits anaerobic digestion, resulting in VFA accumulation and lower biogas production efficiency, although lower levels of FAN were seen in the digesters [29].

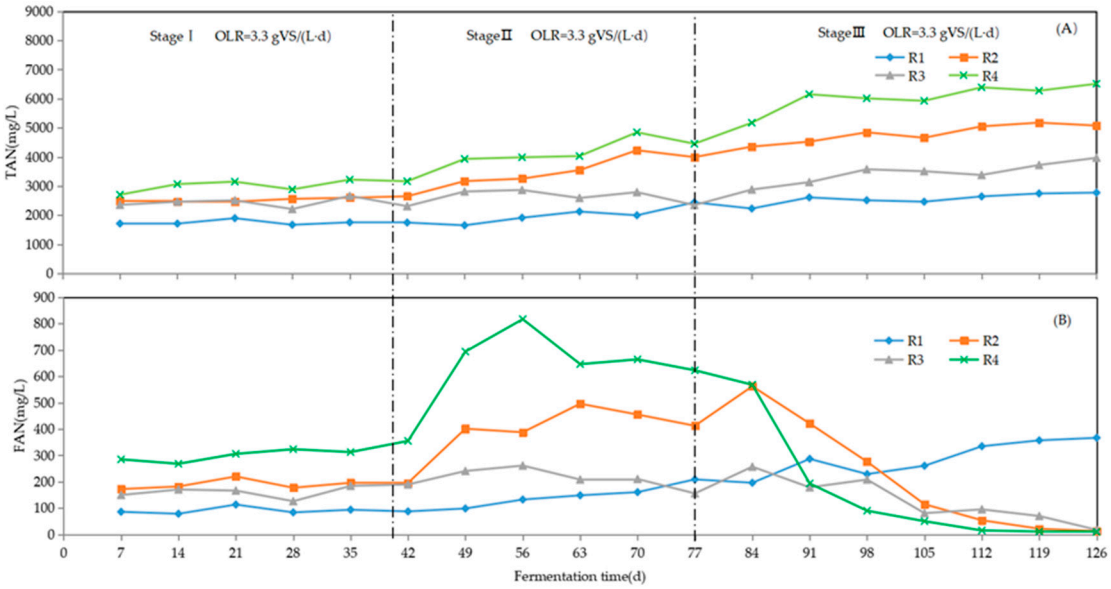


Figure 3. Comparisons of TAN (**A**) and FAN (**B**) between digesters with and without air stripping.

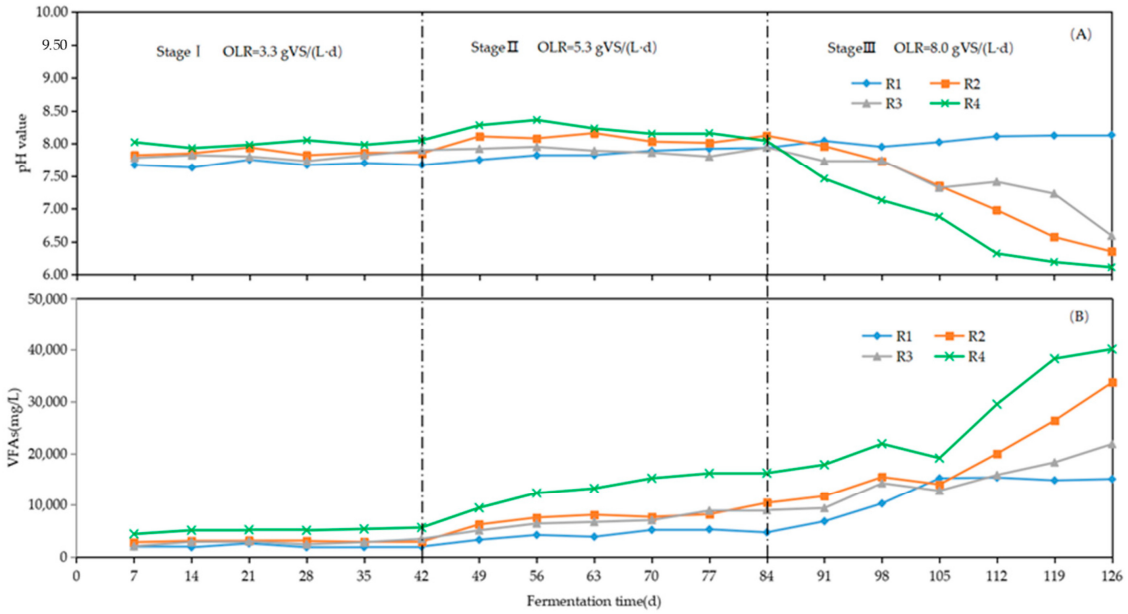


Figure 4. Comparisons of pH (**A**) and VFAs (**B**) between digesters with and without air stripping.

3.2.2. pH and VFAs

As shown in Figure 4, under the same conditions, the amount of VFAs of R1 and R3 (with air stripping) was lower than that of R2 and R4 (without air stripping). VFAs in the digesters with a higher reflux ratio of 75% (R3 and R4) were higher than those at 50% (R1 and R2). At an OLR of 3.3 gVS/(L·d), the VFAs in R1–R3 maintained a lower level, with average values of 2036.96, 2991.29, and 2779.92 mg/L. Relatively stable pH changes were found in R1–R3, within ranges of 7.63–7.75, 7.82–7.94, and 7.73–7.90, respectively. The VFAs and pH of R4 were both higher than those of R1–R3, with average values of 5139.88 mg/L and 8.00. During stage II, the VFAs in R1–R4 gradually increased in response to OLR elevation. On day 84, the VFA concentrations in R2 and R4 increased to 10,395.41 and 16,191.16 mg/L, respectively, while the values in R1 (4714.60 mg/L) and R3 (8997.65 mg/L) were much lower than in R2 and R4. The pH in stage II of R1–R4 also increased as the OLR increased from 3.3 to 5.3 gVS/(L·d), with levels of 7.75–7.93, 8.01–8.16, 7.80–7.95, and 8.04–8.36 recorded, respectively. This increase in pH was mainly due to the adequate

buffer capacity for VFAs attributed to a high TAN concentration (Figure 3A). As the OLR further increased to 8.0 gVS/(L·d), abrupt increases in VFAs were measured in R2–R4, with values of 11,691.85, 9381.74, and 17,812.07 mg/L on day 91, and 33,687.51, 21,816.33, and 40,111.01 mg/L on day 126, respectively, indicating that VFA accumulation occurred. Due to VFA accumulation, the acidity–alkalinity balance of R2–R4 was destroyed. Significant reductions in pH were found in R2–R4, with values of 7.96, 7.73, and 7.46 on day 91, and 6.35, 6.59, and 6.11 on day 126, respectively. Compared with the above digesters, the VFA concentration of R1 was consistently the lowest, with the highest value of 15,039.45 mg/L on day 126. Relatively stable changes in pH during the fermentation process were also found in R1, even at an OLR of 8.0 gVS/(L·d), with a value range of 7.95–8.13.

It can be inferred from the above results that high concentration of VFAs and VFA accumulation occurred in the digesters with direct recycled utilization of the biogas slurry without air stripping, especially under the conditions of a high OLR and reflux ratio. The high concentration of VFAs and VFA accumulation indicated that the activity of the methanogens did not equal the activity of acidogenic bacteria due to the high concentration of TAN and FAN [30]. As mentioned in the above section, although air stripping was conducted, digester R3 was still in the "inhibited steady state" when the OLR was increased to 8.0 gVS/(L·d), indicating that a lower reflux ratio (50%) should be adopted under the conditions of a higher OLR (up to 8.0 gVS/(L·d)).

3.3. VS Removal Efficiency

The VS removal efficiency characterizes the ability of an anaerobic digestion system to treat organic pollutants and, therefore, its value for environmental protection [31]. It can be seen from Figure 5 that the VS removal efficiencies in the digesters were improved by incorporating air stripping. During stages I and II, the R3 digester had a higher VS removal efficiency, with average values of 63.36% and 50.33%, which were 11.53% and 7.89% higher than that of R1, respectively, indicating improvement in the VS removal efficiency via enhancement of the biogas slurry reflux ratio. The improvement of VS removal may have occurred because, with a higher reflux ratio, more microorganisms were returned to the digester to degrade the organic matter [32]. The VS removal efficiency decreased as the OLR increased in the four digesters, especially when the OLR increased to 8.0 gVS/(L·d) when the VS removal efficiencies in R1–R4 decreased to 41.35%, 29.62%, 34.42%, and 25.86%, respectively. The higher OLR may have provided sufficient substrates and also increased the organic matter concentration in the digesters. The organic matter cannot be decomposed in time when the activity of methanogens is inhibited due to a high TAN concentration, resulting in a reduction in VS removal efficiency [30].

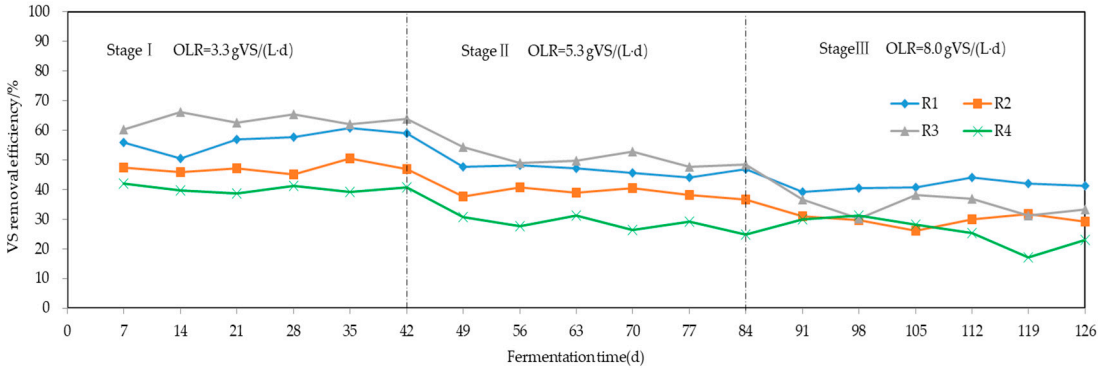


Figure 5. Comparisons of VS removal efficiency between digesters with and without air stripping.

3.4. Microbial Community Analysis

Variations in the relative abundance of methanogens at the genus level are shown in Figure 6. Overall, the methanogenic communities in the four digesters were mainly composed of *Methanosaeta*, *Methanosarcina*, *Methanoculleus*, uncultured_*Bathyarchaeia*, and

Candidatus_Methanoplasma. The total relative abundance values of the five methanogens were in the range of 85.35–93.89% during the fermentation process. *Methanosarcina* and *Methanoculleus* prevailed in the communities of R2 and R4, in which the biogas slurry was returned directly without air stripping. The abundance of *Methanosarcina* decreased and *Methanoculleus* increased as the OLR increased. The methanogen communities of R2 and R4 were dominated by *Methanosarcina*, with a relative abundance of 45.84% and 46.53%, at the OLR of 3.3 gVS/(L·d). *Methanosarcina* is an acetoclastic methanogen that is also capable of producing methane via the hydrogenotrophic pathway [33]. Although its irregular spherical shape means it is adapted to high ammonium levels and unstable fermentation, the high free ammonia and VFA concentration in both R2 and R4 as the OLR increased to a high level (5.3 and 8.0 gVS/(L·d)) still inhibited *Methanosarcina*, reducing biogas production (Figure 2). *Methanoculleus* belongs to the order Methanomicrobiales, which is a hydrogenotrophic methanogen utilizing CO_2 and H_2 as substrates, with a higher tolerance to stress conditions than *Methanosarcina* [19]. Thus, the methanogen communities of R2 and R4 were dominated by *Methanoculleus*, with a relative abundance of 35.82% and 40.68% at an OLR of 5.3 gVS/(L·d), and 41.48% and 46.02% at an OLR of 8.0 gVS/(L·d), respectively. *Candidatus_Methanoplasma* is a methylotrophic methanogen, which increased from 1.70% to 8.40% in R2, and from 1.61% to 9.51% in R4, as the OLR increased from 3.3 to 8.0 gVS/(L·d). Furthermore, after ammonia air stripping, *Methanosaeta* and *Methanosarcina* were the dominant methanogens in R1 and R3. During stage I, a notable amount of *Methanosaeta* (63.01% and 65.72%) was detected in R1 and R3, followed by *Methanosarcina* (8.43% and 7.35%). *Methanosaeta* is generally found in mesophilic conditions and utilizes acetic acid for methane production [34]. As the OLR increased to 5.3 gVS/(L·d), *Methanosaeta* still dominated in R1 and R3, with a relative abundance of 46.92% and 52.63%, respectively, followed by *Methanosarcina* (15.50% and 18.83%). The abundance of *Methanosaeta* showed a downward trend as the OLR increased to 8.0 gVS/(L·d), while *Methanosarcina* exhibited an upward trend in R1 and R3. The increase in *Methanosarcina* was due to its adaptation to higher ammonium from an increasing OLR level (Figure 3).

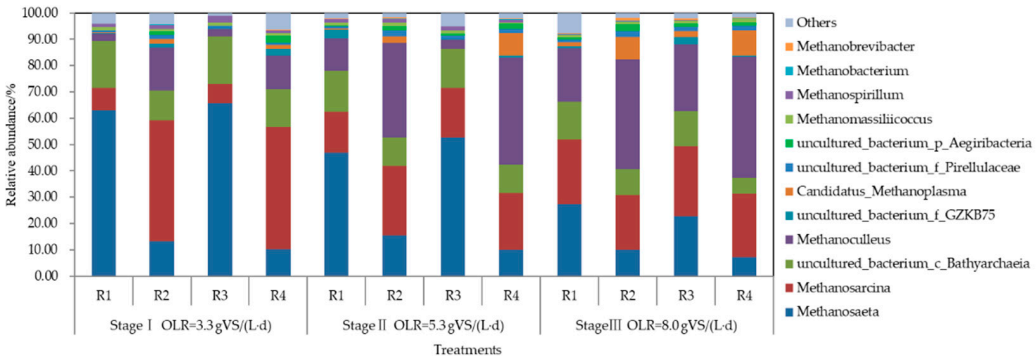


Figure 6. Comparisons and the relative abundance of methanogen communities at the genus level between digesters with and without air stripping (the legend includes genera making up more than 1% of the total composition).

Interestingly, uncultured_*Bathyarchaeia* (phylum Bathyarchaeota) was generally found in the four digesters throughout the fermentation process. Its abundance in R1 (14.41% to 17.87%) and R3 (13.23% to 18.14%) was improved by ammonia air stripping, compared with R2 (9.96% to 11.30%) and R4 (6.00% to 14.37%) without air stripping. It decreased as the OLR increased in the four digesters. *Bathyarchaeia* is considered one of the most abundant archaeal methanogens with complex metabolism pathways and is involved in the fermentation of hemicellulose and cellulose [35–37].

These findings showed that ammonia air stripping improves the structure of methanogen communities and changes the dominant group, with a shift in the methanogenic pathway from hydrogenotrophic to acetoclastic methanogens observed. Previous studies

have reported that acetoclastic methanogens have an advantage over hydrogenotrophic methanogens in biogas production [38], which was consistent with our findings (Figure 2).

4. Conclusions

This work showed that, with air stripping of the digestate, VFA accumulation and ammonium inhibition during the fermentation process were slowed down, and biogas production and VS removal were enhanced. Anaerobic digestion was successfully carried out with air-stripped biogas slurry recirculation at a high OLR (5.3 gVS/(L·d) and reflux ratio (75%), indicating that air stripping enhanced the OLR and reflux ratio. A slightly lower reflux ratio (50%) may be more feasible for a higher OLR (8.0 gVS/(L·d). Furthermore, air stripping changed the dominant methanogen group, with a shift in the methanogenic pathway from hydrogenotrophic to acetoclastic methanogens.

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